

Question

The main component of table salt is NaCl , and the chlorine content in table salt can be determined using the **Mohr** method. The operational steps are as follows:

(1) Accurately weigh 1.3567g of table salt, dissolve it in distilled water, and dilute to 250 mL in a volumetric flask.

(2) Transfer 25.00 mL of this solution to an Erlenmeyer flask, add 6 drops of 50.0 g/L $\text{K}_2\text{Cr}_2\text{O}_7$ solution, and titrate with $0.1000 \text{ mol} \cdot \text{L}^{-1}$ AgNO_3 solution to the endpoint, consuming 22.61 mL. The blank experiment consumed 0.05 mL of AgNO_3 solution.

Assume that the impurity ions in the table salt do not affect the reaction, and assume that $\text{Cr}_2\text{O}_7^{2-}$ is completely converted to CrO_4^{2-} .

$$K_{\text{sp}}(\text{AgCl}) = 1.8 \times 10^{-10}; K_{\text{sp}}(\text{Ag}_2\text{CrO}_4) = 1.2 \times 10^{-12}$$

Then the following statement that is correct is:

- A. All other options are incorrect
- B. The sodium chloride content in table salt is 95.66%.
- C. Assume the titration endpoint is exactly the stoichiometric point. The volume of each drop of 50.0 g/L $\text{K}_2\text{Cr}_2\text{O}_7$ solution is 0.05 mL, at which point Ag_2CrO_4 precipitate forms.
- D. Under the condition of option C, the silver ion concentration in the solution is $8.52 \times 10^{-6} \text{ mol} \cdot \text{L}^{-1}$ at this time.

E. To ensure that the endpoint error of this titration is 0, the volume of indicator solution to be added should be 0.94mL.

F. The method has smaller errors and better results when titrating I^- and SCN^- .

Answer

Correct Answer: **E**

Detailed Explanation

Sodium chloride content calculation:

First, find the amount of AgNO_3 consumed by titration, and then obtain the amount of Cl^- . Combined with solution dilution, extrapolate back to the amount of Cl in the original salt sample, and then calculate the sodium chloride content. The calculation is as follows:

$$\omega = \frac{0.1000 \times (22.61 - 0.05) \times 10^{-3} \times (35.45 + 22.99) \times 10}{1.3567} = 97.18\%$$

CHECKPOINT

1 PTS

Therefore, the sodium chloride content is 97.18%. **B** is incorrect

Assuming that there is only AgCl precipitate and no Ag_2CrO_4 precipitate, then:

$$[\text{Ag}^+] = K_{\text{sp}}^{1/2} = (1.8 \times 10^{-10})^{1/2} = 1.34 \times 10^{-5} (\text{mol} \cdot \text{L}^{-1})$$

CHECKPOINT

1 PTS

Assuming only one precipitate is formed, the calculated silver ion concentration is $1.34 \times 10^{-5} \text{ (mol} \cdot \text{L}^{-1}\text{)}$.

The original concentration of the indicator solution is approximately:

$$[\text{K}_2\text{Cr}_2\text{O}_7] = 0.170 \text{ mol} \cdot \text{L}^{-1}$$

The actual CrO_4^{2-} concentration at the end of the titration is:

$$[\text{CrO}_4^{2-}] = \frac{0.170 \times 2 \times 0.05 \times 6}{25.00 + 22.61 + 0.30} = 2.1 \times 10^{-3} \text{ (mol} \cdot \text{L}^{-1}\text{)}$$

CHECKPOINT

1 PTS

The actual CrO_4^{2-} concentration at the end of the titration is $2.1 \times 10^{-3} \text{ (mol} \cdot \text{L}^{-1}\text{)}$.

To generate Ag_2CrO_4 precipitate, the concentration of CrO_4^{2-} must be at least:

$$[\text{CrO}_4^{2-}] = K_{\text{sp}} / [\text{Ag}^+]^2 = 6.6 \times 10^{-3} \text{ mol} \cdot \text{L}^{-1}$$

CHECKPOINT

1 PTS

Therefore, no precipitate is formed.

$$[\text{Ag}^+] = 1.34 \times 10^{-5} \text{ mol} \cdot \text{L}^{-1}$$

CHECKPOINT

1 PTS

C、**D** are incorrect

To make the endpoint error 0, at the titration endpoint, we should have:

$$[\text{CrO}_4^{2-}] = 6.6 \times 10^{-3} \text{ mol} \cdot \text{L}^{-1}$$

Let x mL of titrant be added, then $[\text{CrO}_4^{2-}] = \frac{0.170 \times 2 \times x}{25.00 + 22.61 + x} = 6.6 \times 10^{-3} \text{ mol} \cdot \text{L}^{-1}$

CHECKPOINT

1 PTS

Solving the equation, we can get $x = 0.94$

The volume of indicator solution to be added is 0.94mL.

CHECKPOINT

1 PTS

E is correct

When using the **Mohr** method, AgI and AgSCN precipitates have strong adsorption, making the endpoint color change unclear and the error large.

CHECKPOINT

1 PTS

F is incorrect